

**Problem 1.****Airport data:**(a) What is the  $\alpha$  if the distribution is power law?

$\alpha = 1.612$ , with  $x_{\min} = 1$  route. Maximum-likelihood estimate  $\hat{\alpha} = 1 + n / \sum_i \ln(x_i / x_{\min})$ ,  $x_{\min} = \min_i x_i$ ,  $n = 3409$ .

|                                                |                      |
|------------------------------------------------|----------------------|
| $n = 3409$                                     | number of airports   |
| $x_{\min} = \min_i x_i = 1$                    | smallest route count |
| $S = \sum_{i=1}^{3409} \ln(x_i / 1) = 5569.43$ | sum of log ratios    |
| $n/S = 3409/5569.43 = 0.6121$                  |                      |
| $\hat{\alpha} = 1 + 0.6121 = 1.612$            | estimate             |

(b) What is the  $\lambda$  if the distribution is exponential?

$\lambda = 0.0504$ . MLE  $\hat{\lambda} = 1/\bar{x} = 1/19.845$ .

|                                     |              |
|-------------------------------------|--------------|
| $\sum_i x_i = 67652$                | total routes |
| $\bar{x} = 67652/3409 = 19.845$     | mean         |
| $\hat{\lambda} = 1/19.845 = 0.0504$ | estimate     |

(c) What is  $[a, b]$  if the distribution is uniform?

$[a, b] = [1, 915]$ , the minimum and maximum route counts (MLE).

|                              |               |
|------------------------------|---------------|
| $\hat{a} = \min_i x_i = 1$   | fewest routes |
| $\hat{b} = \max_i x_i = 915$ | most routes   |

(d) What is  $\mu$  and  $\sigma$  if the distribution is normal?

$\mu = 19.845$ ,  $\sigma = 53.506$  (sample mean and sample standard deviation with  $n - 1$ ).

|                                           |                                 |
|-------------------------------------------|---------------------------------|
| $\hat{\mu} = \bar{x} = 19.845$            | mean, as in the exponential fit |
| $\sum_i (x_i - \bar{x})^2 = 9,756,906.22$ | sum of squared deviations       |
| $9,756,906.22 / (3409 - 1) = 2862.94$     | sample variance                 |
| $\hat{\sigma} = \sqrt{2862.94} = 53.506$  | standard deviation              |

(These can be CDF, PDF, or QQplot)

**Note on the axes.** In every chart on linear axes, the routes-per-airport axis is cut at 200 on purpose: 98% of airports have at most 200 routes, and on a full 0–915 axis they would be squeezed into the left edge. The whole distribution, including the tail up to 915 routes and the model samples beyond it, is shown in the log-log chart next to each linear one. The share of the data and of each model sample beyond 200 is printed on the linear panels.

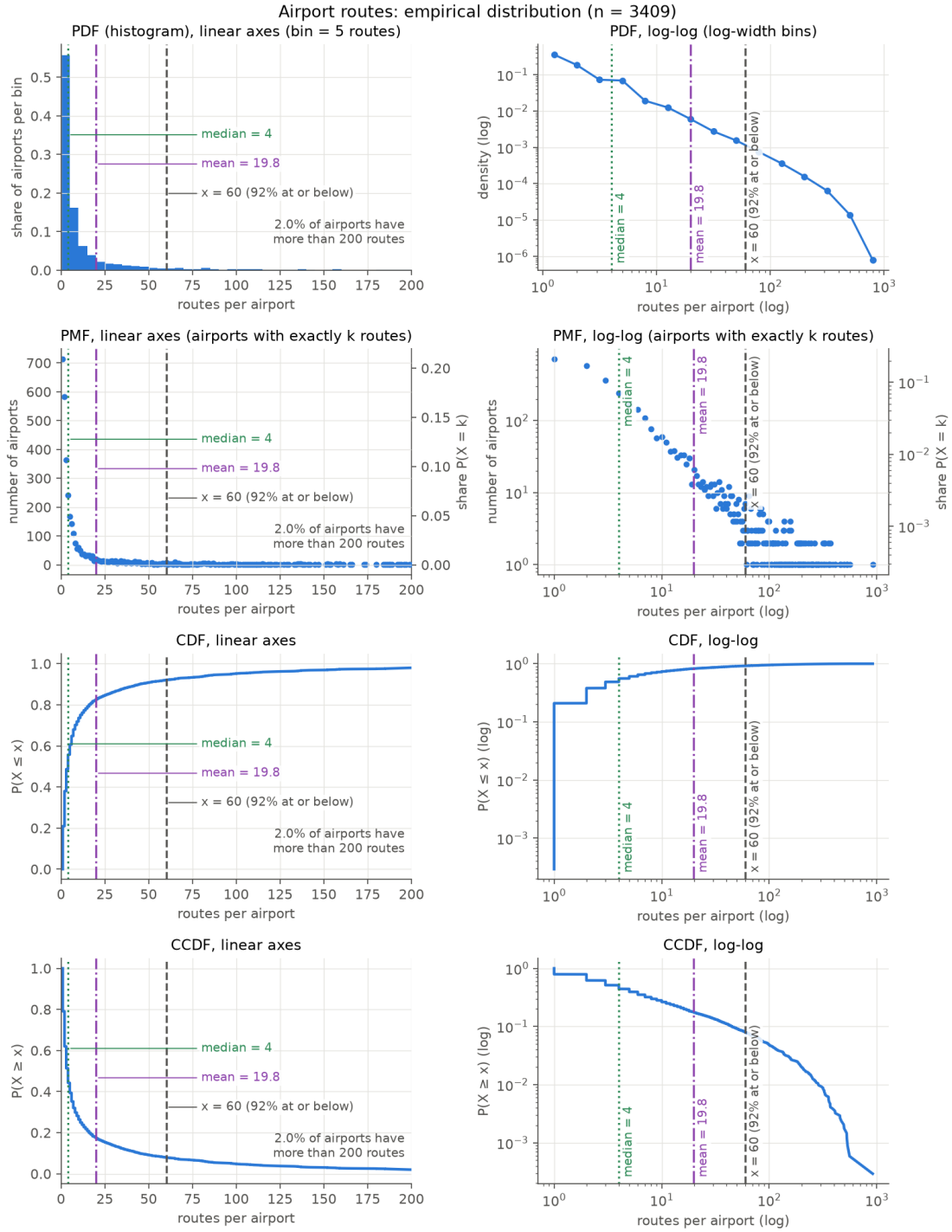
For (e) to (i):

- **Samples.**  $n = 3409$  random values drawn from each fitted model with numpy's random generator (seed 20701313; one generator for all eight samples, so every figure can be rebuilt exactly).
- Power law: numpy has no power-law sampler, so I used inverse transform sampling. Draw  $u$  uniformly from  $[0, 1)$  and set  $x = x_{\min}(1 - u)^{-1/(\alpha-1)}$ , which solves  $u = F(x)$  for the fitted CDF  $F(x) = 1 - (x/x_{\min})^{-(\alpha-1)}$ .
- Exponential: `rng.exponential(1/λ)` (numpy takes the mean  $1/\lambda$ , not  $\lambda$ ).
- Uniform: `rng.uniform(a, b)`. Normal: `rng.normal(μ, σ)`.

The samples are real numbers, not whole route counts: they contain values such as 1.08, and the normal sample contains negative values.

- **Colours.** Data in blue, model sample in orange.
- **Layout of the airport figures.** Rows PDF, CDF, CCDF and QQ, with linear axes on the left and log-log axes on the right; the data's median and mean are marked in every distribution panel, and each CDF/CCDF shows its value at  $x = 1$ .
- **D.** Two-sample Kolmogorov–Smirnov distance: the largest vertical gap between the two CDFs (0 = identical).

(e) Screen shot of the visualization of distribution of data



**Figure 1:** Airport routes: empirical distribution ( $n = 3409$ ). Rows: PDF, PMF, CDF, CCDF. Left column on linear axes, right column on log-log axes.

**PDF, linear (row 1, left).** Histogram with 5-route bins, shown from 0 to 200 routes (2.0% of airports have more than 200); each bar is the share of airports in its bin. 56% of airports fall in the first bin (fewer than 5 routes), 82% have fewer than 20, and the bars beyond about 60 are almost flat on the floor.

**PDF, log-log (row 1, right).** The same distribution with bins of equal width on the log scale, divided by bin width so it is a density. It falls along a roughly straight line and bends down above about 100 routes.

**PMF, linear (row 2, left).** One dot per route count  $k$ : the number of airports with exactly  $k$  routes (left axis) and the same height as a share  $P(X = k)$  (right axis). 713 airports have 1 route (20.9%), 583 have 2, 363 have 3; above about 30 routes every count is at most 14 airports and lies on the floor.

**PMF, log-log (row 2, right).** The same dots on log axes. Up to about 30 routes they fall on a line of slope about  $-1.45$ , close to  $-\hat{\alpha}$ . Right of the dashed line ( $x = 60$ ) each route count belongs to only 1, 2 or 3 airports, so the dots sit in rows at exactly those counts; that is a counting floor, not the shape of the distribution.

**CDF, linear (row 3, left).** The curve rises almost vertically, reaches 0.92 at 60 routes and 0.98 at 200; the last 2% runs on up to the maximum of 915, off this axis.

**CDF, log-log (row 3, right).** The curve starts with a jump to 0.209 at  $x = 1$ , climbs in steps over the first few route counts, and is pressed against 1 from about 20 routes on, so the tail is hardly visible here; that is why the tail is read from the CCDF.

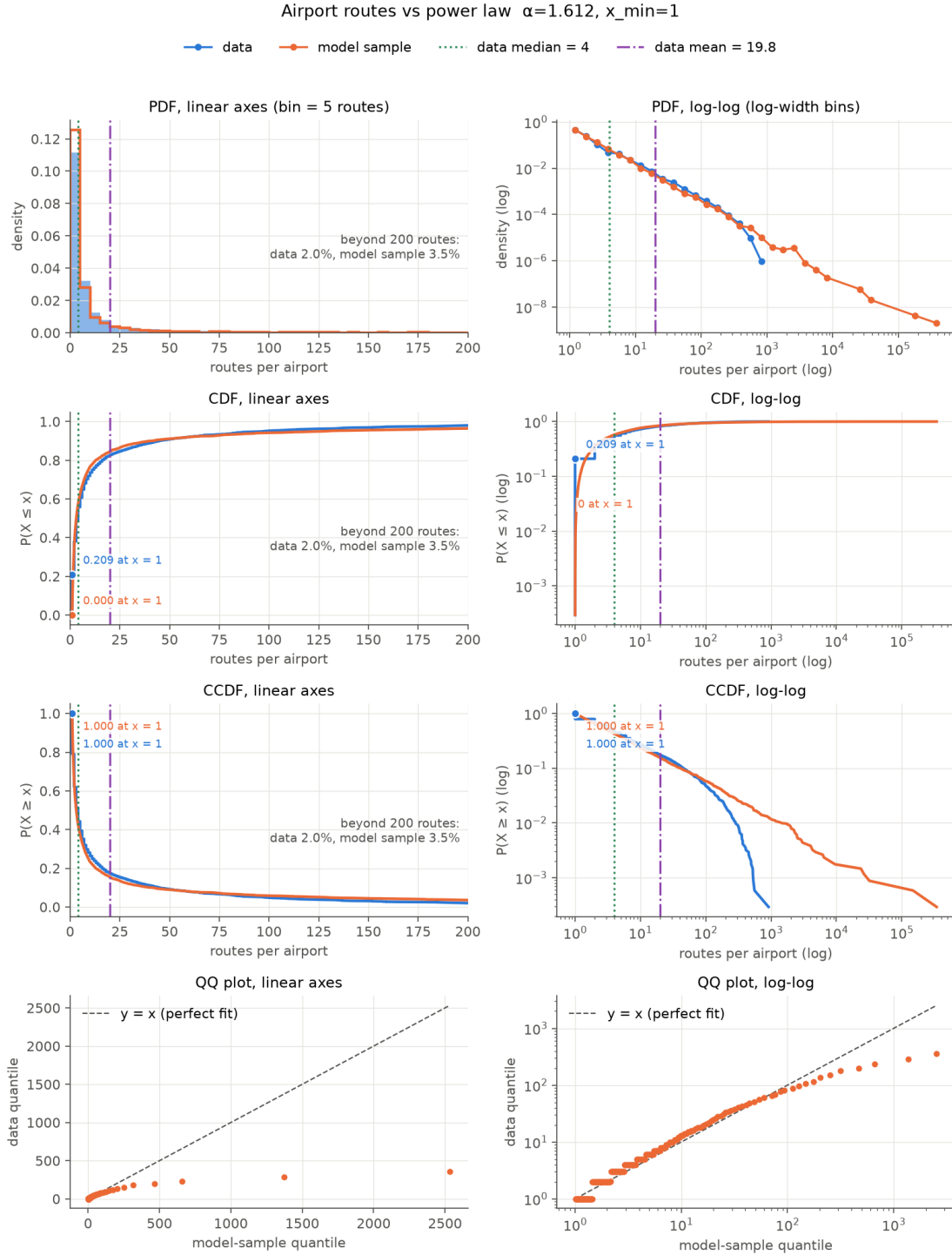
**CCDF, linear (row 4, left).** The share with at least  $x$  routes falls from 1 to 0.18 at 20 routes and 0.08 at 60; from there the tail is a thin line just above zero (0.02 at 200).

**CCDF, log-log (row 4, right).** Share of airports with at least  $x$  routes. It is close to a straight line up to about 100 routes, then bends down and ends at the busiest airport (915 routes).

**Reference lines.** In every panel the green dotted line is the median (4 routes: half the airports have at most 4), the purple dash-dot line is the mean (19.8 routes), and the grey dashed line is  $x = 60$  (92% of airports lie at or left of it). The mean sits five times to the right of the median because a few hubs with hundreds of routes pull it up; on the log-log CCDF only about 18% of airports reach the mean. On linear axes all three lines are squeezed against the left edge.

**Summary.** The median airport has 4 routes and 38% have only 1 or 2, while the busiest has 915. The mean (19.8) is five times the median and the skewness is 6.0, so a few hubs hold most of the routes.

(f) Screen shot of the visualization of distribution of data if it was power law



**Figure 2:** Airport routes vs. power law ( $\alpha = 1.612$ ,  $x_{\min} = 1$ ).  $D = 0.209$ .

**PDF, linear (row 1, left).** Linear panels show 0 to 200 routes. The first 5-route bins are almost the same height for data and model; beyond 200 routes lie 2.0% of the data and 3.5% of the model sample.

**PDF, log-log (row 1, right).** The two lines follow the same straight slope up to about 300 routes. Above that the data drops and ends near 1000, while the model continues past  $10^5$  (its largest sampled airport has 342,563 routes).

**CDF, linear (row 2, left).** The curves almost overlap up to 200 routes. At  $x = 1$  the data is at 0.209

(20.9% of airports have one route) and the model at 0.000: a continuous model puts no mass on exactly one route.

**CDF, log-log (row 2, right).** The log  $x$ -axis spreads out  $x = 1, 2, 3$ , so it shows where the largest gap is: at  $x = 1$ , data 0.209 against model 0.  $D$  itself is measured on the ordinary probability scale,  $0.209 - 0 = 0.209$  (see (j)); from about 3 routes on the curves coincide.

**CCDF, linear (row 3, left).** The curves look identical; on linear axes the tail difference is squeezed against zero. Both start at 1.000 at  $x = 1$ .

**CCDF, log-log (row 3, right).** Both follow the same straight line up to about 100 routes. Above that the data bends down and stops at 915; the model stays straight.

**QQ, linear (row 4, left).** The axis is stretched to the model's 99.5th percentile (2,534 routes), so almost every point is crowded into the bottom-left corner; only the model's few giant airports stand out.

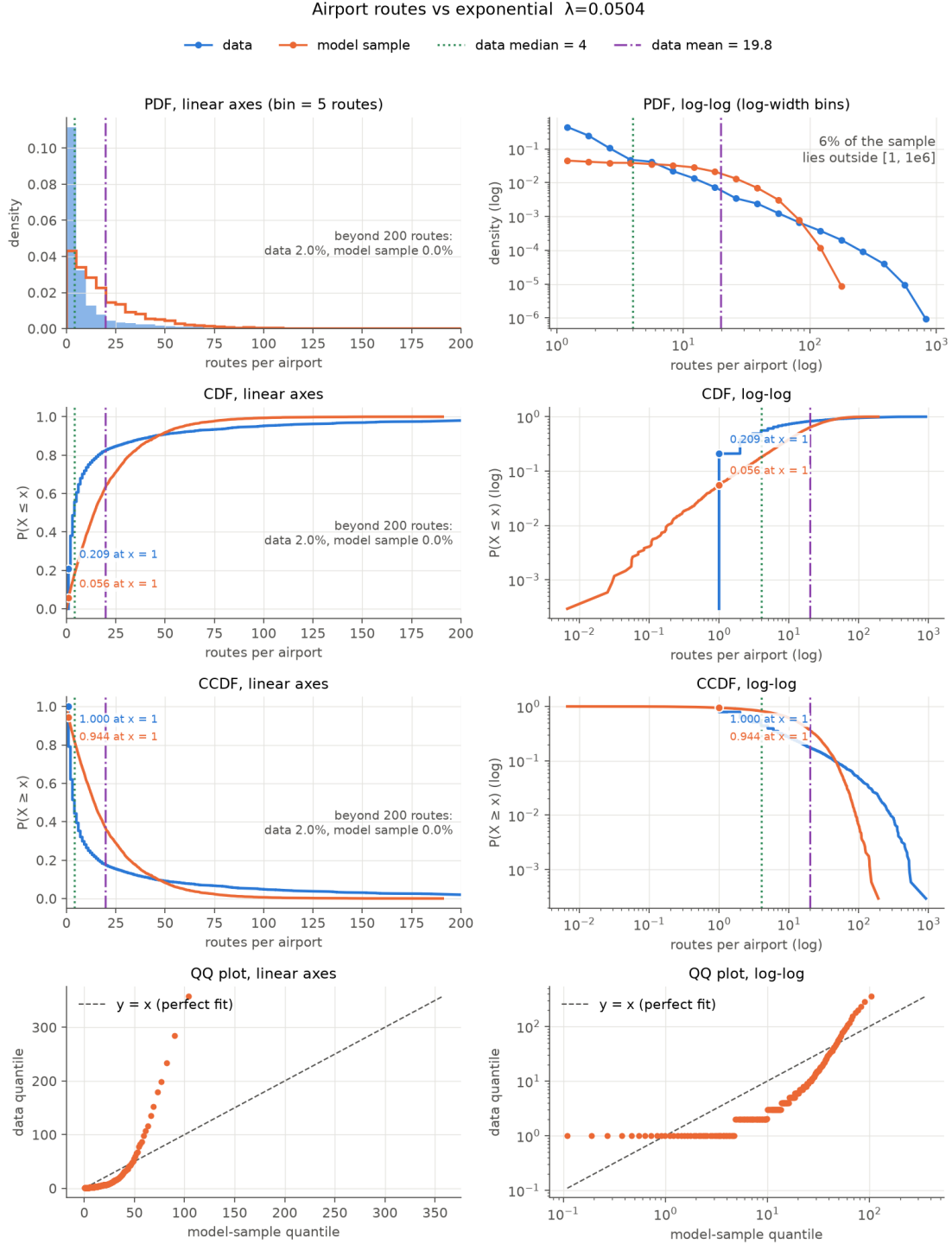
**QQ, log-log (row 4, right).** The points lie on  $y = x$  up to about 60 routes, then run off to the right, below the line: at the 99.5th percentile the model has 2,534 routes and the data 358.

**Why the CCDF on log-log axes.** For a power law, the CCDF is the complement of the CDF, and taking the log of both sides turns it into a straight line:

$$\begin{aligned}
 P(X \geq x) &= 1 - P(X < x) = \left( \frac{x}{x_{\min}} \right)^{-(\alpha-1)} && \text{CCDF of the power law} \\
 \log P(X \geq x) &= -(\alpha - 1) \log x + (\alpha - 1) \log x_{\min} && \text{log of both sides} \\
 \log P(X \geq x) &= -0.612 \log x && \alpha = 1.612, x_{\min} = 1
 \end{aligned}$$

This is linear in  $\log x$  with slope  $-(\alpha - 1) = -0.612$ , so a power law is a straight line on log-log axes and a bend is easy to see. The CDF itself,  $1 - x^{-0.612}$ , keeps the “1–”: its log does not simplify, and on log-log axes it is pressed against 1 in exactly the tail where the model and the data differ.

(g) Screen shot of the visualization of distribution of data if it was exponential



**Figure 3:** Airport routes vs. exponential ( $\lambda = 0.0504$ ).  $D = 0.397$ .

**PDF, linear (row 1, left).** The model's first bin is lower than the data's and it puts more airports between 20 and 100 routes.

**PDF, log-log (row 1, right).** The model is almost flat for small airports and falls off sharply after about 50 routes; the data is steep at the start and has a long, slow tail. 6% of the sample lies below 1 route, which no airport has.

**CDF, linear (row 2, left).** The model reaches 1 by about 190 routes (its largest sampled airport has

191), while the data is only at 0.98 at 200 and keeps rising to 915.

**CDF, log-log (row 2, right).** At  $x = 1$  the data is at 0.209 and the model at 0.056: the model spreads 5.6% of airports below one route and rises smoothly instead of jumping at 1.

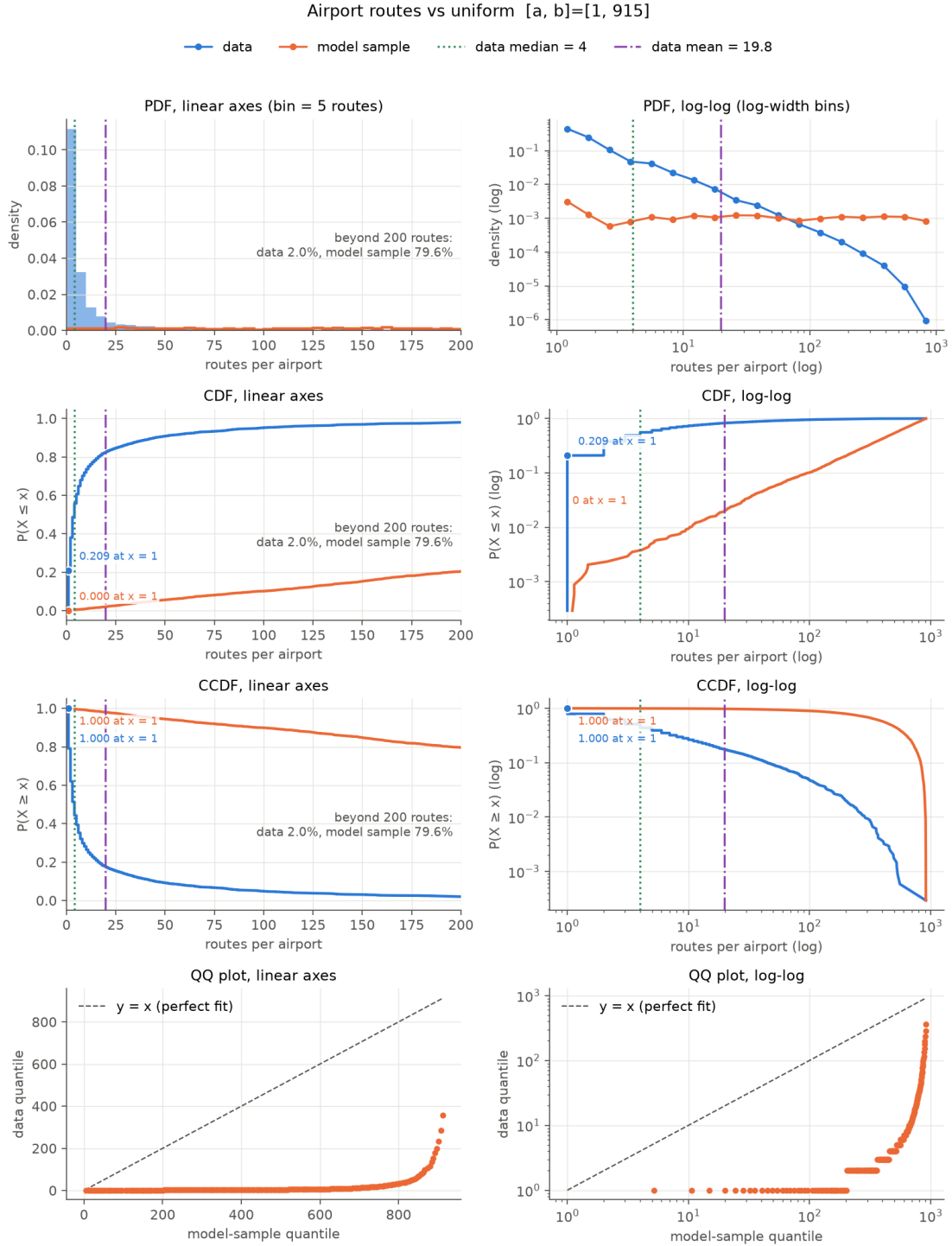
**CCDF, linear (row 3, left).** The model falls more slowly than the data at first, then reaches zero near 190 routes. At  $x = 1$ : data 1.000, model 0.944.

**CCDF, log-log (row 3, right).** The model bends down early and ends near 190 routes; the data goes on to 915. With  $\lambda = 0.05$  an airport with 500 routes has probability about  $e^{-25}$ , yet the data has five such airports.

**QQ, linear (row 4, left).** The points follow  $y = x$  only at the start and then shoot up: at the 99.5th percentile the data has 358 routes and the model about 100.

**QQ, log-log (row 4, right).** The points are too steep: below about 40 routes they lie below  $y = x$  (the model gives small airports too many routes), and above it they rise above the line (hubs get too few).

(h) Screen shot of the visualization of distribution of data if it was uniform



**Figure 4:** Airport routes vs. uniform on  $[1, 915]$ .  $D = 0.856$ .

**PDF, linear (row 1, left).** The model is a flat line at about  $1/914$ ; the data's first 5-route bin is about 100 times higher. 78% of the model sample lies beyond 200 routes, against 2.0% of the data.

**PDF, log-log (row 1, right).** The model stays flat near  $10^{-3}$  across the whole range; the data falls by five orders of magnitude.

**CDF, linear (row 2, left).** The model rises in a straight line and is only at about 0.20 at 200 routes; the data reaches 0.92 by 60 routes. The largest gap between the two,  $D = 0.856$ , is right there, at 60 routes.

**CDF, log-log (row 2, right).** At  $x = 1$  the data is at 0.209 and the model at 0 (the model has no mass at exactly 1); the model stays below 0.1 until about 90 routes.

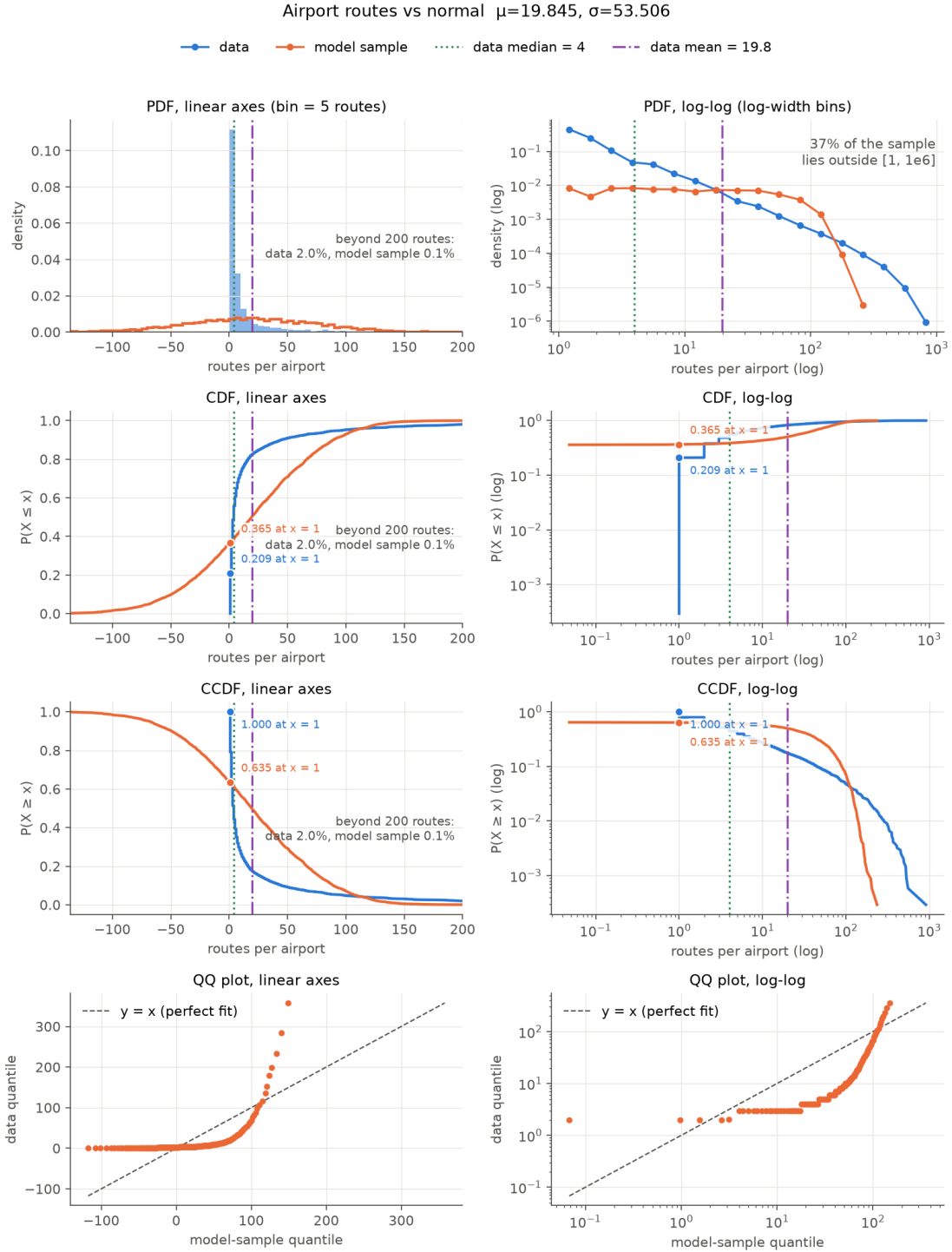
**CCDF, linear (row 3, left).** The model falls in a straight line and is still at about 0.80 at 200 routes, while the data collapses at the left edge. Both are 1.000 at  $x = 1$ .

**CCDF, log-log (row 3, right).** The model stays near 1 until the right edge and then drops vertically at 915.

**QQ, linear (row 4, left).** The points run flat along the bottom: while the model's quantiles climb to about 900 routes, the data's stay near 0 to 20 and rise only at the very end.

**QQ, log-log (row 4, right).** Every point lies far below  $y = x$ , to the right. A typical sampled airport has about 470 routes, while the real median is 4 and 99% of real airports have fewer than 285.

(i) Screen shot of the visualization of distribution of data if it was normal



**Figure 5:** Airport routes vs. normal ( $\mu = 19.8, \sigma = 53.5$ ).  $D = 0.365$ .

**PDF, linear (row 1, left).** The model is a symmetric bell around 20 routes that spreads into negative route counts (down to about  $-136$  at its 0.1st percentile), while the data is one spike at the left edge.

**PDF, log-log (row 1, right).** The model is flat over small airports and then drops steeply after about 100 routes. 37% of its sample is below 1 route, most of it negative, which the log axes cannot show.

**CDF, linear (row 2, left).** The model starts at negative route counts, is at 0.365 at  $x = 1$  against the data's 0.209, and is almost at 1 by 200 routes (its largest sampled airport has 237) while the data goes on

to 915.

**CDF, log-log (row 2, right).** The model is flat at about 0.37 from the left edge to about 10 routes: that plateau is the 37% of the sample below 1 route.

**CCDF, linear (row 3, left).** An S-shaped model curve crossing the data near zero; at  $x = 1$  the model is at 0.635 against the data's 1.000, and it is almost zero by 200 routes.

**CCDF, log-log (row 3, right).** Below about 100 routes the model sits above the data, and its tail ends near 230 routes, far short of 915.

**QQ, linear (row 4, left).** The model's low quantiles are negative (down to about  $-118$ ) while the data's stay at 1 to 3 routes; the points cross  $y = x$  near 100 and then shoot up. The log-log panel cannot show these negative quantiles.

**QQ, log-log (row 4, right).** The points are too steep, as for the exponential: small airports get too many routes and hubs too few. The tail is thin because a normal decays like  $e^{-x^2}$ .

(j) What is the distribution of data based on your observation? Discuss

|                                              | power law    | exponential | uniform | normal |
|----------------------------------------------|--------------|-------------|---------|--------|
| KS distance $D$ (data vs model sample)       | <b>0.209</b> | 0.397       | 0.856   | 0.365  |
| where the gap is largest, $x$                | 1            | 6           | 60      | 0.98   |
| data CDF $F_{\text{data}}(x)$ there          | 0.209        | 0.648       | 0.922   | 0.000  |
| model-sample CDF $F_{\text{model}}(x)$ there | 0.000        | 0.251       | 0.065   | 0.365  |

**How  $D$  is computed.** `scipy.stats.ks_2samp` does not use the figures. It builds the two empirical CDFs from the raw values and, at every  $x$ , measures the gap on the ordinary probability scale;  $D$  is the largest gap:

$$D = \max_x |F_{\text{data}}(x) - F_{\text{model}}(x)|.$$

The axes of a chart do not change  $D$ . A log axis only helps to see where the gap is (for example  $x = 1$  is spread out on a log  $x$ -axis); gap sizes must not be read off a log  $y$ -axis, which stretches small values and cannot show 0. For the power law,  $D = 0.209 - 0.000 = 0.209$  at  $x = 1$ : 20.9% of airports have exactly one route, while the continuous model puts no mass there. The tail mismatch above 100 routes involves few airports, so it barely moves  $D$ ; the figures show it instead. For the normal, the gap sits just below 1, where 36.5% of the model sample already lies (most of it at negative route counts) and no airport does.

**The airport data follows a power law, with a cutoff in the far tail.** It is the only one of the four models whose CCDF is a straight line on log-log axes. The data's CCDF is close to straight for more than two decades (1 to about 100 routes), and the power-law sample matches it there. The other three models fail for structural reasons, not because a parameter is slightly off:

- **Uniform** says every route count from 1 to 915 is equally likely, but most airports are tiny.
- **Normal** is symmetric and allows negative counts, while the data is bounded below at 1 and skewed to the right (skewness 6.0).
- **Exponential** has the right shape (many small, few large), but its tail falls off too fast. With  $\lambda = 0.05$ , an airport with 500 routes has probability about  $e^{-25}$ , yet the data has five airports with more than 500.

The power law is not a perfect fit.  $\hat{\alpha} = 1.61$  is below 2, so the fitted model has an infinite mean and its samples contain airports with  $10^5$  routes. Real airports cannot grow that large: there are only 3409 airports to fly to, and runways, gates and slots limit the busiest hubs. That is why the data's CCDF bends down above about 100 routes. The best description is a power law with an upper cutoff, which is typical of

networks that grow by preferential attachment: new routes go mostly to airports that already have many routes.

## Movie Rating data:

(a) What is the  $\alpha$  if the distribution is power law?

$\alpha = 1.851$ , with  $x_{\min} = 1.9$  (the lowest average vote). Same MLE as for the airports,  $n = 4392$ .

|                                                |                     |
|------------------------------------------------|---------------------|
| $n = 4392$                                     | number of movies    |
| $x_{\min} = \min_i x_i = 1.9$                  | lowest average vote |
| $S = \sum_{i=1}^{4392} \ln(x_i/1.9) = 5163.93$ | sum of log ratios   |
| $n/S = 4392/5163.93 = 0.8505$                  |                     |
| $\hat{\alpha} = 1 + 0.8505 = 1.851$            | estimate            |

(b) What is the  $\lambda$  if the distribution is exponential?

$\lambda = 0.1606$ . MLE  $\hat{\lambda} = 1/\bar{x} = 1/6.227$ .

|                                    |              |
|------------------------------------|--------------|
| $\sum_i x_i = 27348.7$             | sum of votes |
| $\bar{x} = 27348.7/4392 = 6.227$   | mean         |
| $\hat{\lambda} = 1/6.227 = 0.1606$ | estimate     |

(c) What is  $[a, b]$  if the distribution is uniform?

$[a, b] = [1.9, 8.5]$ , the minimum and maximum votes (MLE).

|                              |              |
|------------------------------|--------------|
| $\hat{a} = \min_i x_i = 1.9$ | lowest vote  |
| $\hat{b} = \max_i x_i = 8.5$ | highest vote |

(d) What is  $\mu$  and  $\sigma$  if the distribution is normal?

$\mu = 6.227$ ,  $\sigma = 0.893$  (sample mean and sample standard deviation with  $n - 1$ ).

|                                        |                                 |
|----------------------------------------|---------------------------------|
| $\hat{\mu} = \bar{x} = 6.227$          | mean, as in the exponential fit |
| $\sum_i (x_i - \bar{x})^2 = 3503.28$   | sum of squared deviations       |
| $3503.28/(4392 - 1) = 0.7978$          | sample variance                 |
| $\hat{\sigma} = \sqrt{0.7978} = 0.893$ | standard deviation              |

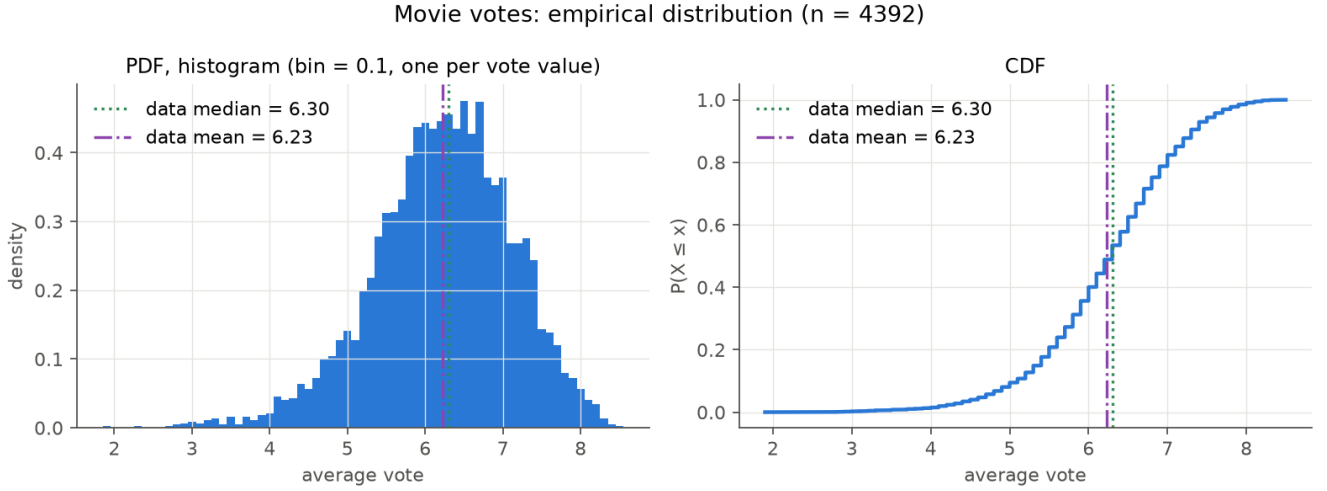
(These can be CDF, PDF, or QQplot)

**Note on the axes.** Average votes lie between 1.9 and 8.5, so every chart uses linear axes. In the model charts (f) to (i) the vote axis is cut at 20 on purpose: the power-law and exponential samples run to values that are not possible ratings (the power law's 99.5th percentile is about 900), and a wider axis would flatten the data's peak. The share of each model sample beyond the axis is printed on the panel, and those QQ plots use log axes so the whole sample is visible.

For (f) to (i):

- **Samples.**  $n = 4392$  random values drawn from each fitted model in the same way as for the airports (same generator, continuing after the airport samples). The samples are not rounded to 0.1 like the votes.

- **Colours.** Data in blue, model sample in orange.
  - **D.** Two-sample Kolmogorov–Smirnov distance: the largest vertical gap between the two CDFs (0 = identical).
- (e) Screen shot of the visualization of distribution of data



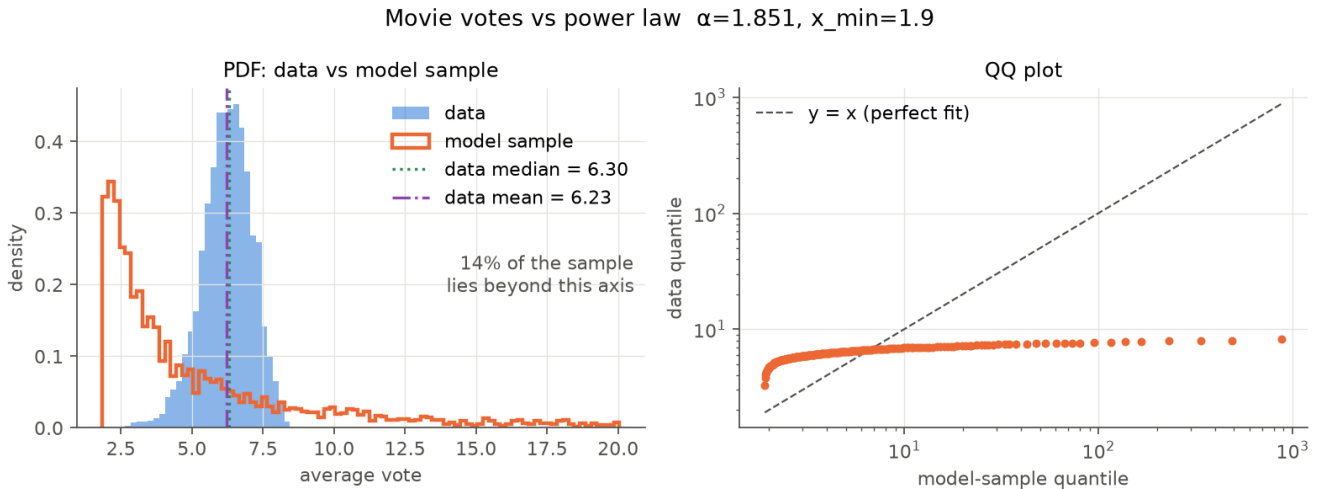
**Figure 6:** Movie votes: empirical distribution ( $n = 4392$ ).

**PDF (left).** Histogram with one bin per possible vote (votes are rounded to 0.1). One peak near 6.3 and a nearly symmetric shape; the left tail is longer (skewness  $-0.48$ ): a few movies are rated 2 to 4, and none are above 8.5.

**CDF (right).** An S-shaped curve centred near 6.3 (median 6.3, mean 6.23). The small steps come from the 0.1 rounding.

**Reference lines.** In both panels (and in the PDF panel of (f) to (i)) the green dotted line is the data median (6.30) and the purple dash-dot line the data mean (6.23). The mean sits just left of the median, as expected for a slightly left-skewed sample: a few very low ratings pull the mean down.

- (f) Screen shot of the visualization of distribution of data if it was power law

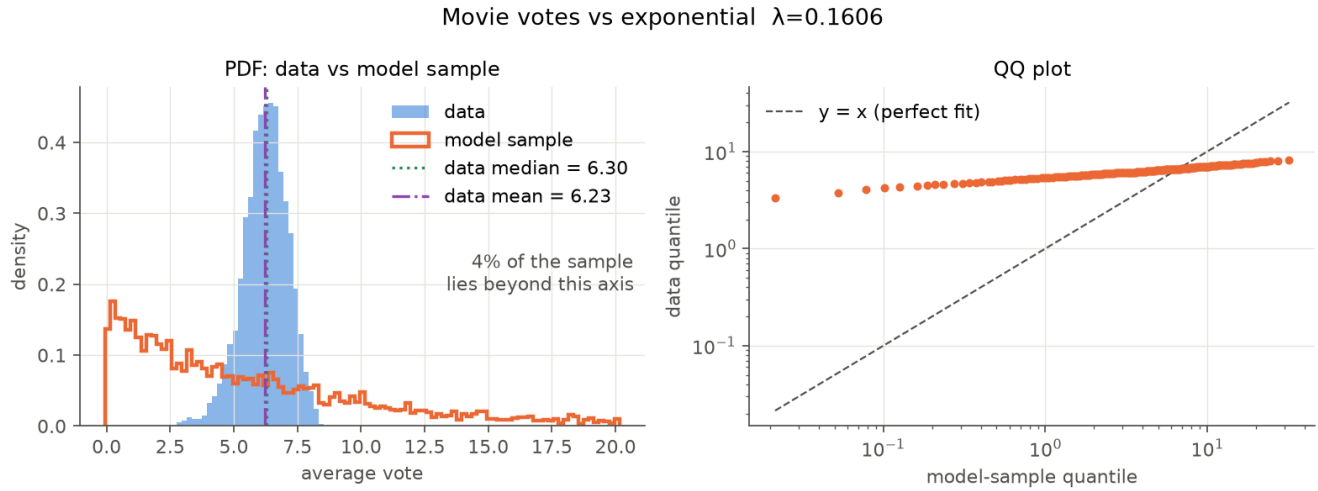


**Figure 7:** Movie votes vs. power law ( $\alpha = 1.851$ ,  $x_{\min} = 1.9$ ).  $D = 0.490$ .

**PDF (left).** The model puts its highest density at the lowest vote (1.9), where the data has almost nothing, and then a long tail. 14% of its sample lies above 20, which is not a possible rating.

**QQ plot (right).** Log axes, because the sample's 99.5th percentile is about 900. At the top votes the points run almost horizontally to the right: the model's high quantiles are hundreds, the data's stop at 8.5.

(g) Screen shot of the visualization of distribution of data if it was exponential

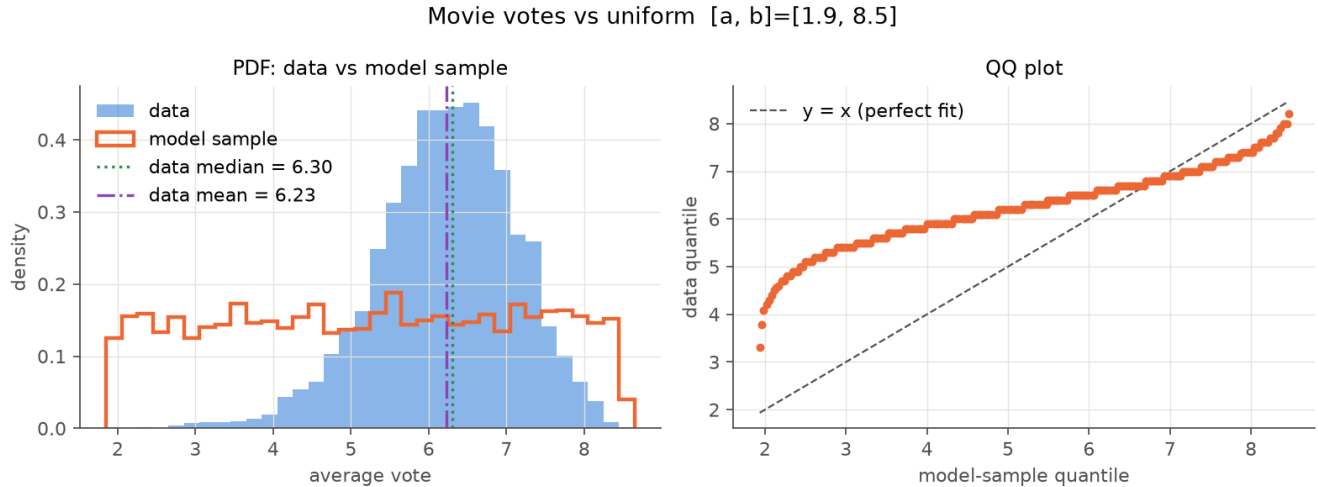


**Figure 8:** Movie votes vs. exponential ( $\lambda = 0.1606$ ).  $D = 0.481$ .

**PDF (left).** The model peaks at zero and has a long right tail; 4% of its sample is above 20. The data is a narrow hump between 4 and 8.5.

**QQ plot (right).** The points cross  $y = x$  at a shallow angle: the model's spread is far larger than the data's, with low quantiles near 0 and high quantiles above 20.

(h) Screen shot of the visualization of distribution of data if it was uniform

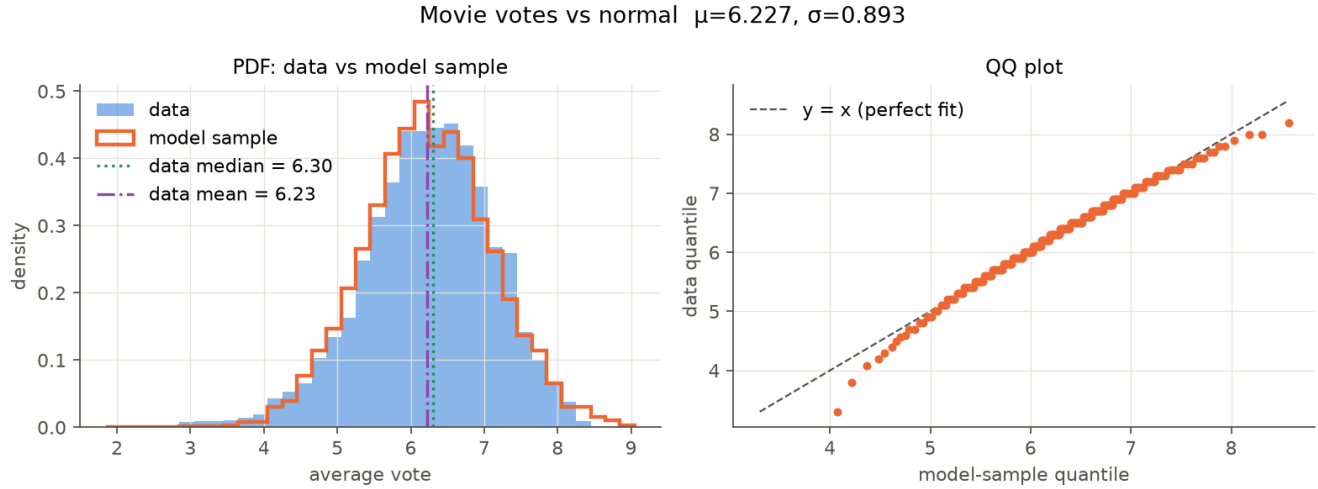


**Figure 9:** Movie votes vs. uniform on  $[1.9, 8.5]$ .  $D = 0.380$ .

**PDF (left).** The model is flat across  $[1.9, 8.5]$ , but the data is concentrated between 5 and 7.5.

**QQ plot (right).** The points lie flat across the middle, above  $y = x$  on the left and below it on the right: the model's low quantiles are too low and its high quantiles too high, because it spreads the votes evenly instead of piling them in the middle.

(i) Screen shot of the visualization of distribution of data if it was normal



**Figure 10:** Movie votes vs. normal ( $\mu = 6.227, \sigma = 0.893$ ).  $D = 0.048$ .

**PDF (left).** The sample traces the same bell as the data, with the peak in the same place.

**QQ plot (right).** The points lie on  $y = x$  from about 4.5 to 7.5. The only departures are at the ends: the data's lowest votes are lower than the model predicts (longer left tail), and the data stops at 8.5 while the model continues.  $D = 0.048$  is by far the smallest.

(j) What is the distribution of data based on your observation? Discuss

|                                              | power law | exponential | uniform | normal       |
|----------------------------------------------|-----------|-------------|---------|--------------|
| KS distance $D$ (data vs model sample)       | 0.490     | 0.481       | 0.380   | <b>0.048</b> |
| where the gap is largest, $x$                | 4.7       | 4.7         | 5.2     | 6.3          |
| data CDF $F_{\text{data}}(x)$ there          | 0.047     | 0.047       | 0.107   | 0.489        |
| model-sample CDF $F_{\text{model}}(x)$ there | 0.538     | 0.528       | 0.488   | 0.536        |

**How  $D$  is computed.** `scipy.stats.ks_2samp` does not use the figures. It builds the two empirical CDFs from the raw values and, at every  $x$ , measures the gap on the ordinary probability scale;  $D$  is the largest gap:

$$D = \max_x |F_{\text{data}}(x) - F_{\text{model}}(x)|.$$

The axes of a chart do not change  $D$ . A log axis only helps to see where the gap is (for example  $x = 1$  is spread out on a log  $x$ -axis); gap sizes must not be read off a log  $y$ -axis, which stretches small values and cannot show 0. For the power law, at a vote of 4.7 only 4.7% of movies are at or below it, but 53.8% of the model sample is; for the normal the largest gap is only  $0.536 - 0.489 = 0.048$ , at 6.3.

**The movie votes follow a normal distribution, approximately.** The data has one peak near the middle of its range and is close to symmetric. The normal model predicts 68.3% of movies within  $\pm 1\sigma$  and 95.4% within  $\pm 2\sigma$ ; the data has 70.1% and 95.6%. In the QQ plot the points lie on the identity line through the bulk of the data, and the KS distance is about one eighth of the next-best model's.

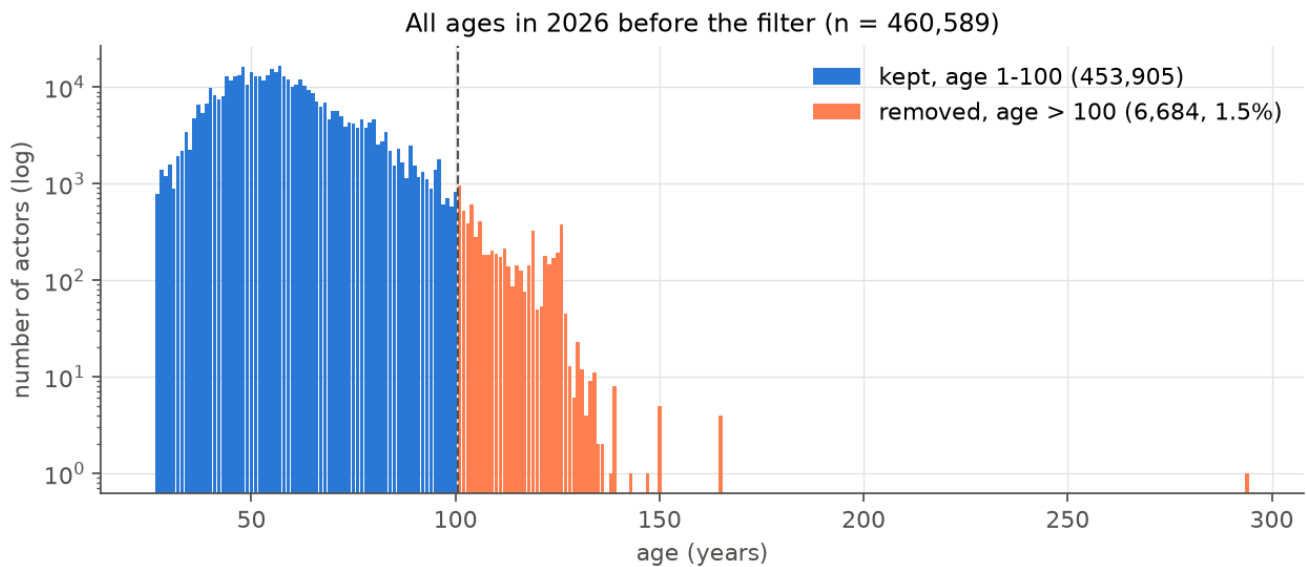
The other three models fail for structural reasons. The power law and the exponential are both monotonically decreasing: their most likely value is the smallest one, so they predict mostly terrible movies and a long tail of ratings above 10. The uniform model ignores the peak altogether.

The fit is good but not exact, for two reasons. First, the data is bounded to  $[1, 10]$  and slightly left-skewed: bad movies can drop further below the average than good movies can rise above it. Second, votes are rounded to one decimal, which gives the QQ plot its small steps. A normal shape is expected because each value is an *average* of many individual votes. By the central limit theorem an average of many independent ratings tends toward a normal distribution, whatever the distribution of each single rating.

## Problem 2.

### - Screenshot of distribution/histogram of the age

How the ages were computed. The birth dates have the form **dd-Mon-yy** (a few use a 4-digit year). Every 2-digit year is read as **19yy**: IMDB-WIKI was collected in 2015, so a year such as 20–24 cannot mean 2020–2024, and the counts per year decrease smoothly from the 1920s down to 00, with no break that would mark a switch to the 2000s. The 129 rows with year 0000 (a missing-value placeholder) were dropped. Per the professor’s instruction (2026-09-23), age is now the year difference **2026 minus birth year** (month/day ignored), kept only if  $1 \leq \text{age} \leq 100$ . Keeping that range leaves **453,905** people (mean 56.87, median 55, range 27 to 100). The minimum of 27 is a result of the **19yy** rule, since the latest possible birth year is 1999 ( $2026 - 1999 = 27$ ). Of the rest, 6,684 rows had age  $> 100$  and were dropped; none had age  $< 1$ .

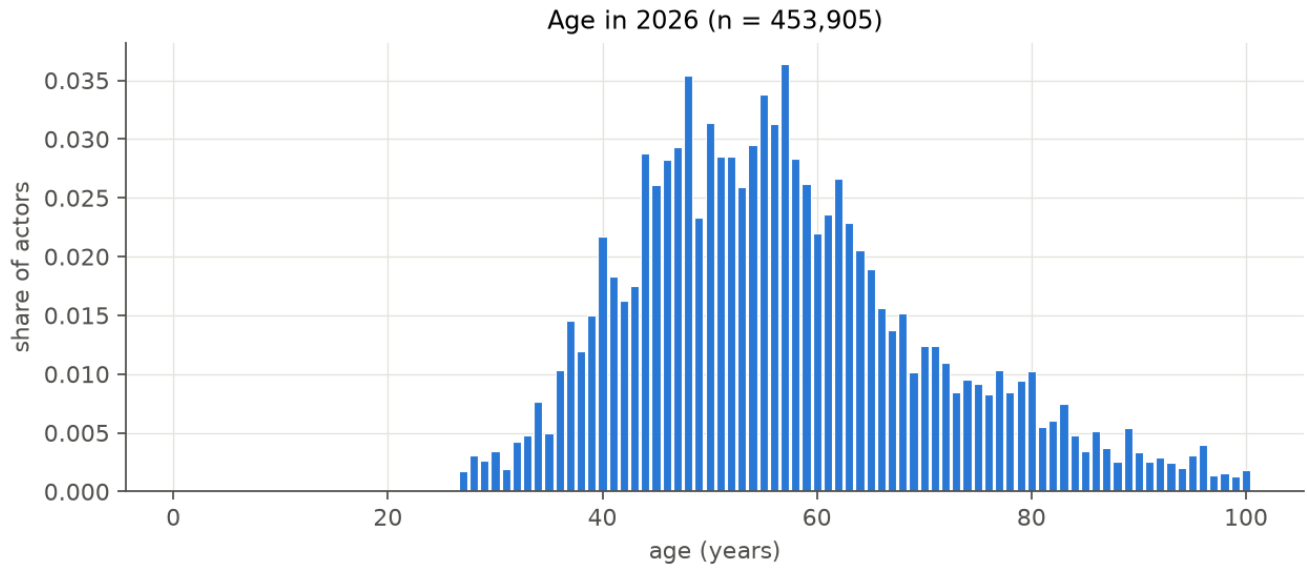


**Figure 11:** All ages in 2026 before the  $1 \leq \text{age} \leq 100$  filter ( $n = 460,589$ ). Blue bars are kept, coral bars are removed; log  $y$  axis.

**Kept (blue).** Ages 27 to 100, 453,905 people. No age is below 27 because the latest birth year is 1999.

**Removed (coral).** Ages 101 to 294, 6,684 people (1.5%). Most are 101 to about 130; a few isolated bars reach 150, 165 and one at 294 (born 1732). These are not plausible ages for people in a 2015 photo database, so the filter drops them. The 129 rows with year 0000 are not ages at all and are dropped before this plot.

**Why log  $y$ .** The removed years hold only 1.5% of people; on a linear axis their bars would be invisible next to the peak of about 16,000 per year.

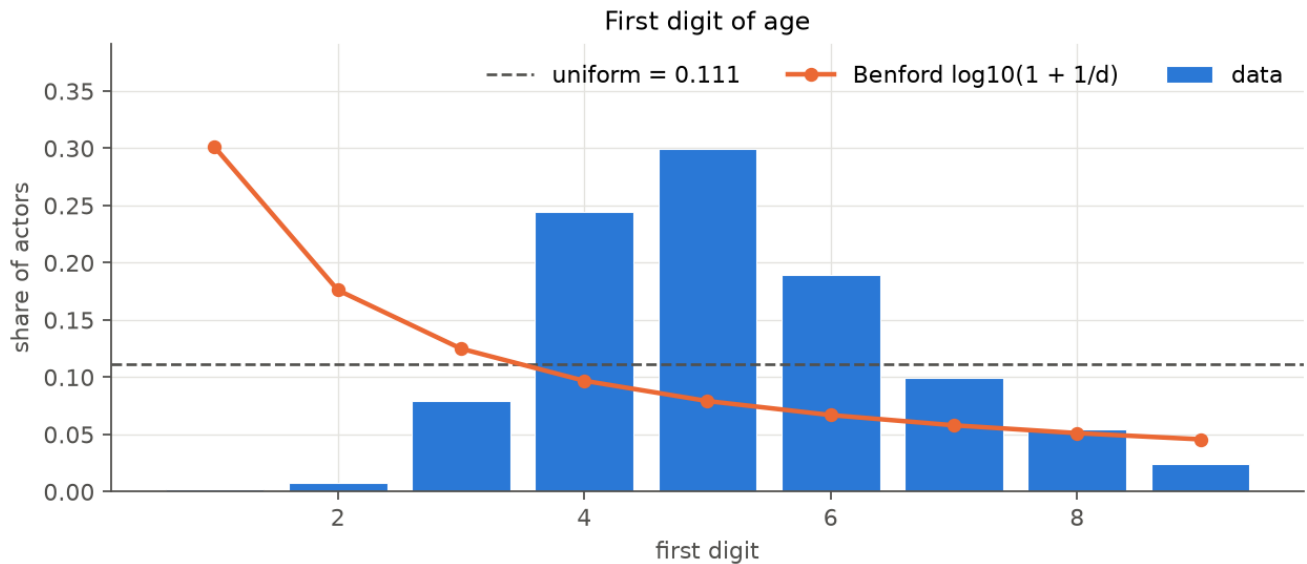


**Figure 12:** Age in 2026 ( $n = 453,905$ ).

**Shape.** One broad hump: ages start at 27, rise through the 30s, peak in the 40s and 50s, and fall off in a long right tail up to 100. Most of the mass is between 36 and 71.

**Jagged bars.** Neighbouring years differ (for example, 48 and 57 stand out). This is year-to-year noise in how many actors were born in a given year, not a pattern in the ages themselves.

- Screenshot of distribution/histogram of the first digit of age



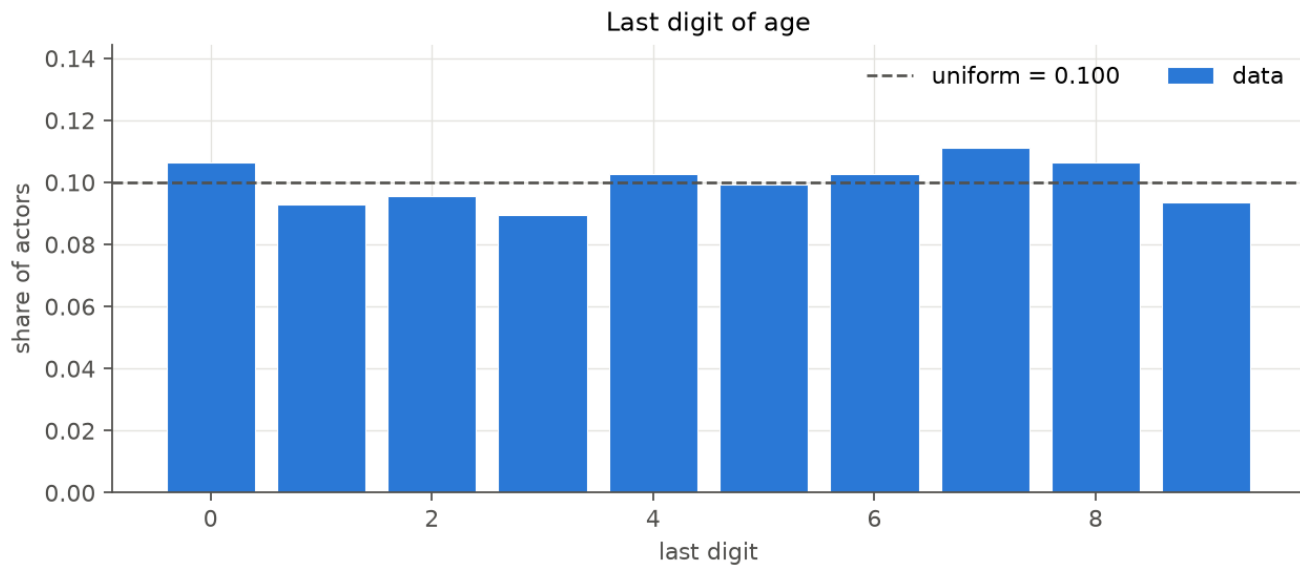
**Figure 13:** First digit of age, with the uniform level  $1/9$  (dashed) and Benford's law (orange).

**Bars.** The shares are 0.2%, 0.7%, 7.9%, 24.5%, 30.0%, 18.9%, 9.9%, 5.4% and 2.5% for digits 1 to 9. They peak at 5 and 4, the same place as the hump in the age histogram.

**Uniform line.** Far from the dashed  $1/9 = 0.111$  level: digits 4 to 6 are well above it, digits 1 to 3 and 8 to 9 well below.

**Benford curve.** Also far from Benford, which predicts 1 as the most common digit (30.1%); here 1 is the rarest.

- Screenshot of distribution/histogram of the last digit of age



**Figure 14:** Last digit of age, with the uniform level  $1/10$  (dashed).

**Bars.** Every digit has a share between 8.95% (digit 3) and 11.10% (digit 7), within 1.10 percentage points of 10%.

**Uniform line.** All ten bars sit close to the dashed 0.100 level, with no digit clearly favoured.

- Discussion (what are those distribution?) Why do you think it is like this?

**The last digit is uniform (approximately); the first digit is not.**

**Last digit.** This is expected. The last digit of an age is its value modulo 10. Every block of ten ages (20–29, 30–39, ...) contains each last digit exactly once. The age distribution changes slowly compared with a step of one year, so within any block of ten the ten ages have roughly equal counts, and summing over the blocks gives each last digit about 10%. The remaining deviations of up to  $\pm 1.1$  points come from year-to-year noise in how many actors were born in a given year (the jagged bars in the age histogram), not from any preference for particular digits.

**First digit.** This is also expected, once the two boundary decades are accounted for separately. For ages 30 to 99, the first digit *is* the decade of life, and grouping by decade reproduces the shape of the age distribution: it peaks at 5 (the 50s, 30.0%) and 4 (the 40s, 24.5%), matching the median of 55. Digit 2 is small (0.7%) not because few actors are in their 20s, but because the  $1 \leq \text{age} \leq 100$  rule only keeps ages 27 to 29 in that decade, three values out of ten, a boundary effect of the reference year rather than a property of the data. Digit 1 is smaller still (0.2%): no age from 10 to 19 survives the rule (the youngest possible age is 27), so the entire digit-1 count comes from the 816 people with age exactly 100 ( $816/453,905 = 0.18\%$ , matching the observed share). For the first digit to be uniform, every decade would need the same number of actors, which working actors clearly are not, and the two extreme digits are further skewed by where the  $[1, 100]$  window happens to cut.

**It is not Benford's law either.** Benford's law (first digit  $d$  with probability  $\log_{10}(1 + 1/d)$ , so 1 is the most common at 30.1%) applies to data that spans several orders of magnitude, such as populations, prices or the airport route counts in Problem 1, where the log of the value is spread out roughly evenly. Age spans less than one order of magnitude (27 to 100) and is concentrated around 55, so its first digit shows where that one hump sits, not Benford's pattern. In short, the last digit is uniform because it throws away where

the value is and keeps only its position within a block of ten. The first digit keeps exactly that information, so its distribution follows the shape of the age distribution.

**Tooling.** Python 3 / pandas / numpy / scipy / matplotlib; all estimates computed programmatically, none typed by hand; analysis code, figure code and write-up drafted with Claude Code. Code: one script per answer in `hw2/code/` (for example `p1_airport_a_alpha.py`), sharing `common.py` and `common_p2.py`; `run_all.py` runs them all.